

PH.48111

FALL 2026

Autonomous Theory Construction Laboratory

Course Syllabus

Garam Yun

Fall 2026

Instructor	Garam Yun
Teaching assistants	Saebom Lee, Minjae Gu
Semester	Fall 2026
Language	English

Course description

Students design and evaluate iterative research systems using explicit data schemas, candidate grammars, objective functions, holdout data, and reproducible computational environments.

Learning outcomes

- Specify a machine-readable scientific objective and reference data set.
- Implement a constrained grammar for candidate models.
- Separate proposal, evaluation, validation, and holdout stages.
- Produce a reproducible research release with complete run records.

Assessment

Component	Weight
Weekly laboratories	30%
Midterm reproduction	20%
Peer comparison	15%
Final release	25%
Participation	10%

References

- G. Yun, Autonomous Theory Construction: Registry, Loop, Release, lab manual rev. 31.
- Research Tool Attribution Guide MAS-54 (2026 consolidated ed.).

- Autonomous Research Bureau, Run History Protocol RLP-12.
- Campus Theory-Refinement Archive, preserved regression snapshot CRS-26-SAT.

Weekly schedule

Week	Dates	Topic
1	2026-08-31–2026-09-06	Iterative Research Systems
2	2026-09-07–2026-09-13	Designing Reference Data
3	2026-09-14–2026-09-20	Grammar of Candidate Theories
4	2026-09-21–2026-09-27	Attribution and Reproducibility
5	2026-09-28–2026-10-04	Multi-Objective Loss and Covariance
6	2026-10-05–2026-10-11	Auxiliary Terms and Local Gates
7	2026-10-12–2026-10-18	Run Histories and Reproduction
8	2026-10-19–2026-10-25	Reproducing a Converged Search
9	2026-10-26–2026-11-01	Automated Manuscript Generation
10	2026-11-02–2026-11-08	Validation Sets and Negative Controls
11	2026-11-09–2026-11-15	Cross-Domain Theory Grammars
12	2026-11-16–2026-11-22	Peer Comparison
13	2026-11-23–2026-11-29	Holdout Evaluation
14	2026-11-30–2026-12-06	Preparing a Research Release
15	2026-12-07–2026-12-13	Final Holdout

Submission policy

Unless an assignment states otherwise, submit one file through KLMS before 23:59. Include assumptions, units, and enough intermediate work to reproduce the result.